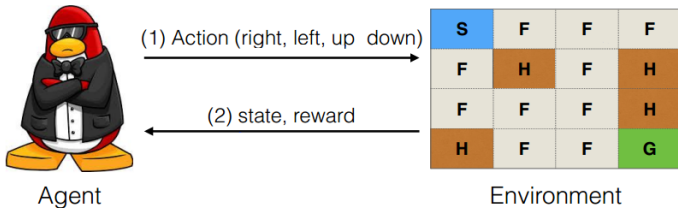


Lecture 4: Model-Free Reinforcement Learning: Monte-Carlo, SARSA, Q-Learning

Anton Plaksin

Frozen Lake World (OpenAI GYM)



Example: Breakout Atari Game



- States: pixels
- Actions: $\rightarrow$, $\leftarrow$, «0»
- Rewards: points
- Final states: when the ball falls down

Markov Decision Process

Markov Property

$$\mathbb{P}[S_{t+1}|S_t, A_t] = \mathbb{P}[S_{t+1}|S_1, A_1, S_2, A_2 \dots, S_t, A_t]$$

$$\mathbb{P}[R_t|S_t, A_t] = \mathbb{P}[R_t|S_1, A_1, S_2, A_2 \dots, S_t, A_t] = 1$$

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Markov Decision Process $\langle \mathcal{S}, \mathcal{S}_F, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

- $\mathcal{S}$ is a **finite** ($|\mathcal{S}| = n$) state space
- $\mathcal{S}_F$ is a set of final states
- $\mathcal{A}$ is a **finite** ($|\mathcal{A}| = m$) action space
- $\mathcal{P}$ is an **unknown** transition probability function

$$\mathcal{P}(s'|s, a) = \mathbb{P}[S_{t+1} = s'|S_t = s, A_t = a]$$

- $\mathcal{P}_0$ is an **unknown** initial state probability function
- $\mathcal{R}$ is an **unknown** reward function

$$\mathcal{R}(s, a) = R_t \quad \Leftrightarrow \quad \mathbb{P}[R_t|S_t = s, A_t = a] = 1$$

- $\gamma \in [0, 1]$ is a discount coefficient

Model-Free Algorithms

- Monte-Carlo Algorithm
- SARSA Algorithm
- Q-Learning Algorithm

Policy Iteration

Let π^0 and $L, K \in \mathbb{N}$

For each $k \in \overline{0, K}$, do

- (Policy evaluation) Iterative Policy Evaluation:

$$v^{l+1}(s) = \sum_a \pi(a|s) \left(\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v^l(s') \right), \quad l \in \overline{0, L-1}.$$

Define $q^L(s, a)$ by $v^L(s)$

- (Policy improvement) Greedy Policy Improvement:

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} q^L(s, a') \\ 0, & \text{otherwise} \end{cases}$$

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to find $q_\pi(s, a) = \mathbb{E}_\pi[G \mid S_0 = s, A_0 = a] \approx Q(s, a)$

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$$Q_N = \frac{1}{N} \sum_{i=1}^N w_i$$

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Monte-Carlo Policy Evaluation

Let π be fixed. Set $Q(s, a) = 0$ and $N(s, a) = 0$.

For each $k \in \overline{1, K}$, do

- According to π , get a trajectory $\tau = (S_0, A_0, \dots, S_T)$ and rewards $(R_0, \dots, R_{T-1})$. Define $(G_0, \dots, G_{T-1})$.

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Monte-Carlo Policy Evaluation

Let π be fixed. Set $Q(s, a) = 0$ and $N(s, a) = 0$.

For each $k \in \overline{1, K}$, do

- According to π , get a trajectory $\tau = (S_0, A_0, \dots, S_T)$ and rewards $(R_0, \dots, R_{T-1})$. Define $(G_0, \dots, G_{T-1})$.
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$$Q(s, a) \approx q_\pi(s, a)$$

Will it work?

Let π^0 be initialized and $K > 0$.

For each $k \in \overline{1, K}$, do

- (Policy evaluation) Monte-Carlo Policy Evaluation.
Obtain $Q^k(s, a) \approx q_{\pi^k}(s, a)$
- (Policy improvement) Greedy Policy Improvement:

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q^k(s, a') \\ 0, & \text{otherwise} \end{cases}$$

Multi-Armed Bandit



- States: the start and final states
- Actions: $\rightarrow$, $\leftarrow$
- Rewards:
 $R(S_0, \leftarrow) = 1$,
 $R(S_0, \rightarrow) = 2$
- Initial Policy:
 $\pi^0(S_0, \leftarrow) = 1$,
 $\pi^0(S_0, \rightarrow) = 0$
- $Q^0(S_0, \leftarrow) = ?$,
 $Q^0(S_0, \rightarrow) = ?$

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- Initial Policy:
 $\pi^0(S_0, \leftarrow) = 1$,
 $\pi^0(S_0, \rightarrow) = 0$
- $Q^0(S_0, \leftarrow) = 1$,
 $Q^0(S_0, \rightarrow) = 0$
- $\pi^1 = \pi^0$,

Multi-Armed Bandit



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- $\pi^1 = \pi^0$,
- $Q^1 = Q^0$,

ε -Greedy Policy Improvement

$$\pi = \varepsilon\text{-greedy}(Q)$$

$$\pi(a|s) = \begin{cases} 1 - \varepsilon + \varepsilon/m, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q(s, a'), \\ \varepsilon/m, & \text{otherwise} \end{cases}$$

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Policy Improvement Theorem

Let $Q(s, a)$ be defined.

Let $\pi = \varepsilon\text{-greedy}(Q)$ and $\pi' = \varepsilon\text{-greedy}(q_\pi)$.

Then $\pi' \geq \pi$ ($v_{\pi'}(s) \geq v_\pi(s)$, $\forall s$)

Learning with Monte-Carlo Policy Evaluation

Let π^0 be initialized and $K > 0$, $\varepsilon = 1$.

For each $k \in \overline{1, K}$, do

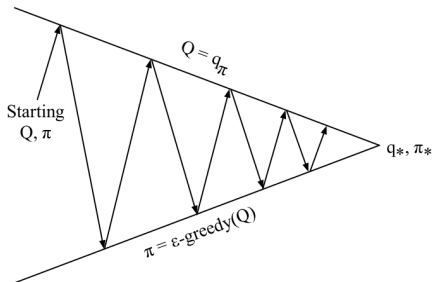
- (Policy evaluation) Monte-Carlo Policy Evaluation - obtain $Q^k(s, a) \approx q_{\pi^k}(s, a)$
- (Policy improvement) ε -Greedy Policy Improvement - obtain π^{k+1} by Q^k . Define $\varepsilon = 1/k$

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Theorem

$Q^k \rightarrow q_*$ and $\pi^k \rightarrow \pi_*$ as $k \rightarrow \infty$.

Monte-Carlo Algorithm

Let $Q(s, a) = 0$, $N(s, a) = 0$ and $\varepsilon = 1$.

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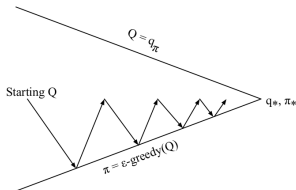
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Теорема

$Q^k \rightarrow q_*$ and $\pi^k \rightarrow \pi_*$ as $k \rightarrow \infty$.

Using Bellman Equation

Bellman Expectation Equation для q_π

$$q_\pi(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) \sum_{a'} \pi(a'|s') q_\pi(s', a')$$

$\Downarrow$

$$q_\pi(s, a) = \mathbb{E}_\pi[R_t + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]$$

$\Downarrow$

Temporal-Difference

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t))$$

Temporal-Difference Policy Evaluation

Let π be fixed and $Q(s, a) = 0$.

For each $k \in \overline{1, K}$, do

- According to π , get trajectory $\tau = (S_0, A_0, \dots, S_T)$ and rewards $(R_0, \dots, R_{T-1})$.

Temporal-Difference Policy Evaluation

Let π be fixed and $Q(s, a) = 0$.

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- According to π , get trajectory $\tau = (S_0, A_0, \dots, S_T)$ and rewards $(R_0, \dots, R_{T-1})$.
- For each $t \in \overline{0, T-1}$, update Q :

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$$Q(s, a) \approx q_\pi(s, a)$$

Comparison of MC and TD Policy Evaluation

Two states A , B ; no discounting; 8 episodes of experience

$A, 0, B, 0$

$B, 1$

$B, 1$

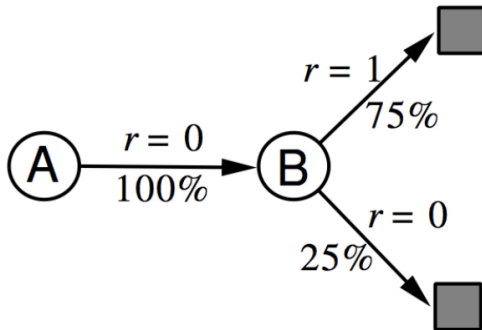
$B, 1$

$B, 1$

$B, 1$

$B, 1$

$B, 0$



What is $V(A)$, $V(B)$?

Learning with Temporal-Difference Policy Evaluation

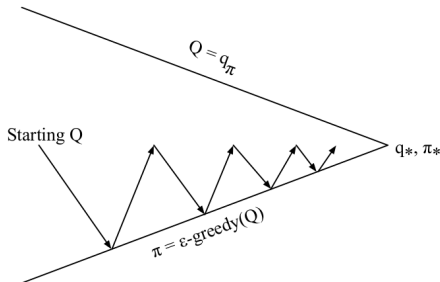
Let $Q(s, a) = 0$, $K > 0$, and $\varepsilon = 1$.

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Define $\varepsilon = 1/k$



SARSA Algorithm

Let $Q(s, a) = 0$, $K > 0$, and $\varepsilon = 1$.

For each $k \in \overline{1, K}$, do

During trajectory

- From the state S_t , acting $A_t \sim \pi(\cdot|S_t)$, where $\pi = \varepsilon$ -greedy(Q), get R_t , go to the next state S_{t+1} , and act $A_{t+1} \sim \pi(\cdot|S_{t+1})$
- According to $(S_t, A_t, R_t, S_{t+1}, A_{t+1})$, update Q :

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Using Bellman Optimality Equation

Bellman Optimality Equation для q_*

$$q_*(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) \max_{a'} q_*(s', a')$$

$\Downarrow$

$$q_*(s, a) = \mathbb{E}[R_t + \gamma \max_{a'} q_*(S_{t+1}, a') \mid S_t = s, A_t = a]$$

$\Downarrow$

Q-Learning

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(R_t + \gamma \max_{a'} Q(S_{t+1}, a') - Q(S_t, A_t))$$

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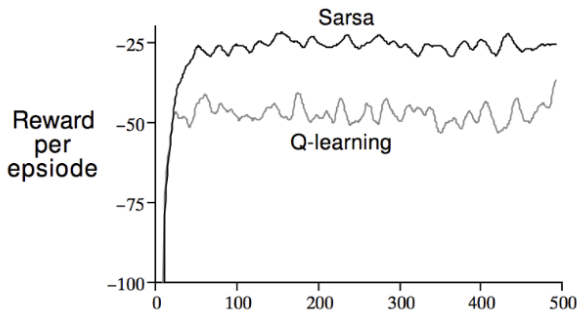
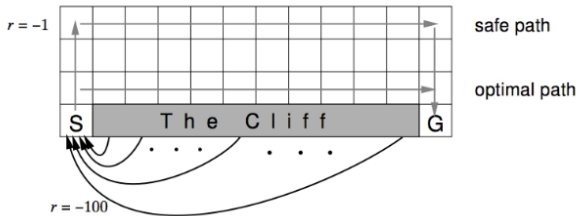
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$Q^k \rightarrow q_*$ and $\pi^k \rightarrow \pi_*$ as $k \rightarrow \infty$.

Comparison of SARSA и Q-Learning



Dynamic Programming and Reinforcement Learning

Q-Policy Iteration

$$Q(s, a) \leftarrow \mathbb{E}[R + \gamma Q(S', A') \mid s, a]$$

Sarsa

$$Q(S, A) \stackrel{\alpha}{\leftarrow} R + \gamma Q(S', A')$$

Q-Value Iteration

$$Q(s, a) \leftarrow \mathbb{E} \left[R + \gamma \max_{a' \in \mathcal{A}} Q(S', a') \mid s, a \right]$$

Q-Learning

$$Q(S, A) \stackrel{\alpha}{\leftarrow} R + \gamma \max_{a' \in \mathcal{A}} Q(S', a')$$

QUESTIONS?